

Blake Allen Linders

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Education

Texas A&M University

Bachelor of Science in Electrical Engineering, Minor in Math – GPA: 3.33

August 2024 – Present

(Expected May 2028)

- Data Analysis Certificate – Katzberg Institute
- IEEE Member and Multiple Intramural Sports

Skills

Hardware/Lab Tools: FPGA, Breadboard, Oscilloscope, DAQ, Multimeters

Software/Simulation: MATLAB, Simulink, LTspice, Vivado

Programming Languages: Python (NumPy, SymPy), PyTorch, C++, Java, Verilog (Behavioral and Structural)

Engineering Concepts: Digital Systems Design, Circuit Theory, Machine Learning, Data Analysis, Finite State Machines, Op-Amps, Logic Gates

Projects

Voltage Controlled Oscillator Analysis

- Constructed a physical VCO circuit on a breadboard, measuring and analyzing output frequency variations across a continuous 1V to 8V input voltage sweep.

FPGA Traffic Light Controller System

- Engineered an FSM using Verilog to manage complex timing and state transitions for a multi-directional traffic intersection.
- Synthesized logic using Vivado and deployed the state machine to a ZYBO Z7-10 FPGA board while integrating clock dividers to translate high-frequency system clocks into real-time physical delays.

FPGA Combination Lock System

- Designed and deployed multi-state FSM to validate sequential multi-bit user inputs.
- Programmed precise state transitions triggered by tactile button inputs on the ZYBO board to verify 3 and 4-digit sequences.
- Mapped hardware signals on a breadboard to drive physical LED status indicators for unlock confirmation.

Diode & LED I/V Characterization

- Conducted physical laboratory experiments to measure, graph, and analyze I/V curves with an oscilloscope of standard diodes and green LEDs.
- Developed accurate piecewise linear models for the components by calculating empirical voltage drops across resistors.

Machine Learning & Data Analysis

- Analyzed complex neural network architectures, calculating VGG16 CNN weights and mapping decision tree boundaries to evaluate predictive model performance.
- Executed K-means clustering iterations using Python and utilized Business Intelligence frameworks to extract, process, and analyze actionable insights from raw data.

Experience & Leadership

Blue Stars Drum and Bugle Corps

Section Leader (Bassline)

- Outstanding First Year Battery Member Award

May 2025 – August 2025

Lewisville High School

Drumline Technician

August 2024 – April 2025

Pacific Crest Youth Arts Organization

Performing Member (Bassline)

May 2024 – August 2024